



Automatic detection and measurement system for aortic aneurysms using deep learning-based artificial intelligence

Jumpei Fujiwara¹ · Makoto Orii¹ · Kohei Oyamada² · Kenta Araki² · Masahiko Ogura² · Tomoyasu Ito² · Daria Bunu² · Ryoto Ando² · Takashi Okada² · Kunihiro Yoshioka¹

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Abstract

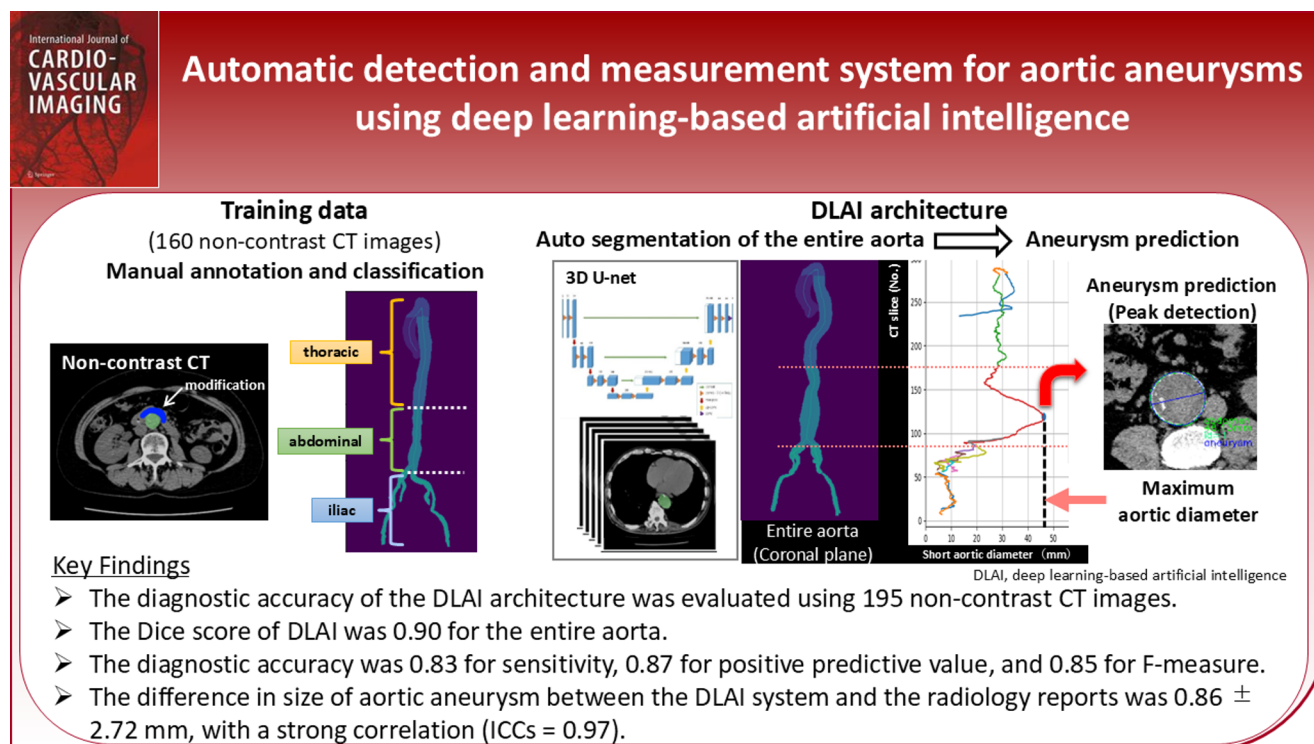
This study evaluated the performance of a deep learning-based artificial intelligence (DLAI) system for detecting fusiform aortic aneurysms and measuring the diameters of the entire aorta on non-contrast CT images. We retrospectively collected 160 non-contrast CT images (130 males, 71.8 ± 9.2 years) to evaluate aortic aneurysms and used them as training data. To evaluate the accuracy of the DLAI system for the automatic detection and measurement of aortic aneurysms, another 190 non-contrast CT images (153 males, 70.7 ± 10.3 years) were collected and used as validation data. First, the accuracy of DLAI results was calculated by referring to the radiology reports. An expert radiologist read the test data again and calculated the final accuracy by referring to the DLAI predictions. Dice scores for aortic segmentation were 0.90, 0.94, 0.93, and 0.84 for the entire aorta, thoracic aorta, abdominal aorta, and iliac artery, respectively. The sensitivity, positive predictive value, and F-measure of the DLAI system for detecting aortic aneurysms were 0.81, 0.83, and 0.82, respectively. Eleven (4.4%) aneurysms were newly identified after the radiologist's second reading based on DLAI predictions. The final accuracy of the DLAI detection increased to 0.83, 0.87, and 0.85 for sensitivity, positive predictive value, and F-measure, respectively. The size error in the measurement of aortic aneurysms between the DLAI system and radiology reports was 0.86 ± 2.72 mm, with a strong correlation ($\text{ICCs} = 0.97$). Our DLAI system was highly accurate in detecting aneurysms and measuring the diameters of the entire aorta on non-contrast CT images.

✉ Makoto Orii
kori931@gmail.com

¹ Department of Radiology, Iwate Medical University, Yahaba, Morioka, Japan

² Technology and Innovation General HQs, NTT DATA Group Corporation, Toyosu, Tokyo, Japan

Graphical Abstract



Keywords Aortic aneurysm · Deep learning-based artificial intelligence · Non-contrast computed tomography

Introduction

Aortic aneurysms are frequently asymptomatic but potentially life-threatening conditions. Aortic aneurysms with higher diameters are associated with increased risks and are typically detected incidentally on imaging studies [1–3]. Therefore, high-quality aortic aneurysm imaging is essential for accurate assessment and clinical decision-making.

The detection and measurement of aortic aneurysms on computed tomography (CT) are recommended by several guidelines [1, 2]. Although CT angiography (CTA) with dedicated workstations allows three-dimensional (3D) assessment of the aorta, expertise and precise manual analysis are required, which is labor-intensive and time-consuming and often complicated by intricate aortic anatomy. The inter-observer reproducibility of aortic shape measurement is poor at 87%, exceeding the clinical tolerance of ± 5 mm for aortic diameter measurement [4, 5].

Recently, deep learning-based artificial intelligence (DLAI) models, particularly the U-Net architecture, have made significant advances in medical image segmentation [6]. Advancements in DLAI have enabled more accurate segmentation of various medical conditions and internal structures of the human body [7–9]. Several studies have

reported the efficacy of CTA-based fully automated segmentation and measurement of aortic aneurysms using DLAI methods [10–13]. However, no studies have examined the fully automated detection and measurement of aneurysms of the entire aorta based on non-contrast CT images using DLAI methods. Thus, this study estimated the diagnostic accuracy of a DLAI system for detecting and measuring the diameter of fusiform aneurysms of the entire aorta in non-contrast CT images.

Methods

Cohort selection

The cohort comprised two datasets of non-contrast CT images that included either the thoracic or abdominal (abdominal and iliac region), and these were acquired retrospectively from the picture archiving and communication system (PACS) of Iwate Medical University Hospital (Table 1). The study protocol conformed to the ethical guidelines of the 1975 Declaration of Helsinki, as reflected in an a priori approval by the institution's human research committee (MH2021-135). The requirement for informed

Table 1. Baseline characteristics of Datasets A and B

	Dataset A	Dataset B
Number of studies	160	190
Dates captured	2011/5–2018/1	2011/5–2018/5
Age, years	71.8±9.2	70.7±10.3
Male, n (%)	130 (81)	153 (81)
Protocol		
Chest CT, n (%)	9 (6)	25 (13)
Chest to abdomen CT, n (%)	89 (56)	116 (61)
Abdominal CT, n (%)	62 (39)	49 (26)
Agatston score		
Thoracic	4576 [1475–14155]	3778 [779–11860]
Abdomen	2659 [1201–6265]	2622 [1036–6871]
Iliac	2749 [1175–7037]	2695 [843–7089]
Section thickness		
1–3 mm, n (%)	89 (56)	75 (39)
4–6 mm, n (%)	71 (44)	115 (61)
Diagnosis according to radiology reports		
Thoracic aorta aneurysms, n (%)	39 (24)	81 (33)
Abdominal aorta aneurysms, n (%)	89 (54)	127 (52)
Iliac artery aneurysms, n (%)	36 (22)	36 (15)
Total, n	164	244

consent for this retrospective analysis was waived by the Institutional Review Board.

Training dataset

Dataset A was gathered for model development and testing using a postprocessing algorithm. The data included 160 non-contrast CT images (130 men, mean age 71.8±9.2 years) obtained between May 2011 and January 2018 for the evaluation of aortic aneurysms or dissections. The slice thickness of the images ranged from 1 to 5 mm. Of all images, nine were chest CT images, 62 were abdominal CT images, and 89 were chest plus abdominal CT images. Based on the clinical definition, the presence of fusiform aneurysms was determined by applying ≥ 45 mm for the thoracic, ≥ 30 mm for the abdominal, and ≥ 20 mm for the iliac regions to the maximum aortic diameter as defined by manual segmentations [1]. Based on the radiology reports, the maximum aortic diameter and slice location of each aortic aneurysm were documented to generate training data [1]. When multiple aneurysms were present in a single CT image, they were counted separately. Thus, 39 thoracic, 89 abdominal, and 36 iliac artery aneurysms were observed in the 160 non-contrast-enhanced CT images (Table 1).

Manual ground truth segmentation and regional annotation

To generate a ground truth aortic segmentation, the entire aorta was manually contoured on the axial scans slice-by-slice between the aortic root and bilateral common femoral artery. Each study was annotated by two radiologists (F.J: 5 years of experience in cardiovascular imaging research and O.M: 7 years of experience in cardiovascular imaging research). Irfan View (<https://www.irfanview.com/>) and 3D Slicer (<https://www.slicer.org>) were used to delineate and modify the manual annotations. In the case of smaller vessels, such as the iliac artery, and imaging with thicker slice thicknesses, fragmented annotations were observed. To address this issue, a 3D Slicer was used to perform 3D annotations in the axial, coronal, and sagittal planes. Furthermore, regional annotations were conducted for the thoracic, abdominal, and iliac regions. The separation of the thoracic and abdominal regions is located above the celiac trunk. The boundaries of the abdominal and iliac regions were located just above the bifurcation of the left and right iliac arteries. For the final set of weights, the complete primary dataset was randomly split into training (80%), validation (10%), and test (10%) sets.

Validation dataset

Dataset B was used to assess the model against radiologist-reported ground truth and to validate the postprocessing stages that generate the presence and measurement of the aneurysms. The data included 190 non-contrast CT images (153 men, mean age 70.7±10.3 years) obtained between May 2011 and May 2018 to evaluate aortic aneurysms. The slice thickness of the images ranged from 1 to 6 mm. The images included 25 chest, 49 abdominal (abdominal and iliac region), and 116 chest plus abdominal CT images. In these images, 81 thoracic, 127 abdominal, and 36 iliac artery aneurysms (IAAs) were observed.

Both datasets were acquired using a 320 multidetector row CT scanner (Canon Medical Systems, Otawara, Japan). No distinction was made between aneurysms with and without dissection, and no studies were excluded based on image quality to reflect the heterogeneity of studies observed in routine clinical practice. Data were de-identified at the time of extraction. Calcification in the thoracic, abdominal, and iliac regions was quantitatively assessed in both the training and validation datasets using the Agatston method (Table 1) [14].

Automatic aneurysm detection pipeline overview

Fusiform aortic aneurysms were detected and measured using the following methods (Fig. 1): (a) transfer of the Digital Imaging and Communications in Medicine (DICOM) data to the segmentation model based on the 3D U-Net architecture; (b) automatic volume segmentation of the entire aorta and site segmentation; (c) the aortic centerline determined using 3D skeletonization and drawing the perpendicular view to the centerline; and (d) measurement of the minimum aortic diameter in each vertical cross-sectional slice and detection of aneurysms using a peak detection algorithm.

Segmentation model architecture and training

A multiclass segmentation model was constructed to infer semantic segmentation of the entire aorta and annotated aortic regions (thoracic, abdominal, and iliac regions) using the

manual annotated dataset as training data (Figs. 1a and 1b). The segmentation model was developed using the Medical Open Network for Artificial Intelligence (<https://monai.io/>), based on the 3D U-Net architecture [15, 16]. The layers included 3D Convolution, 3D Transposed Convolution, Parametric Rectified Linear Unit activations, and 3D batch normalization. The Dice similarity coefficient was used as the loss function, and optimization during training was performed using the Adam algorithm. Closing operations were employed to remove the noise encapsulated within the segmentation.

Postprocessing and aneurysm detection algorithms

The postprocessing pipeline comprised several algorithmic tasks (Figs. 1c and 1d). First, the aortic centerline was determined by 3D skeletonization. Centerlines with distances below a threshold were removed to extract the main aortic centerline (removing outlier). Based on the algorithm, a line

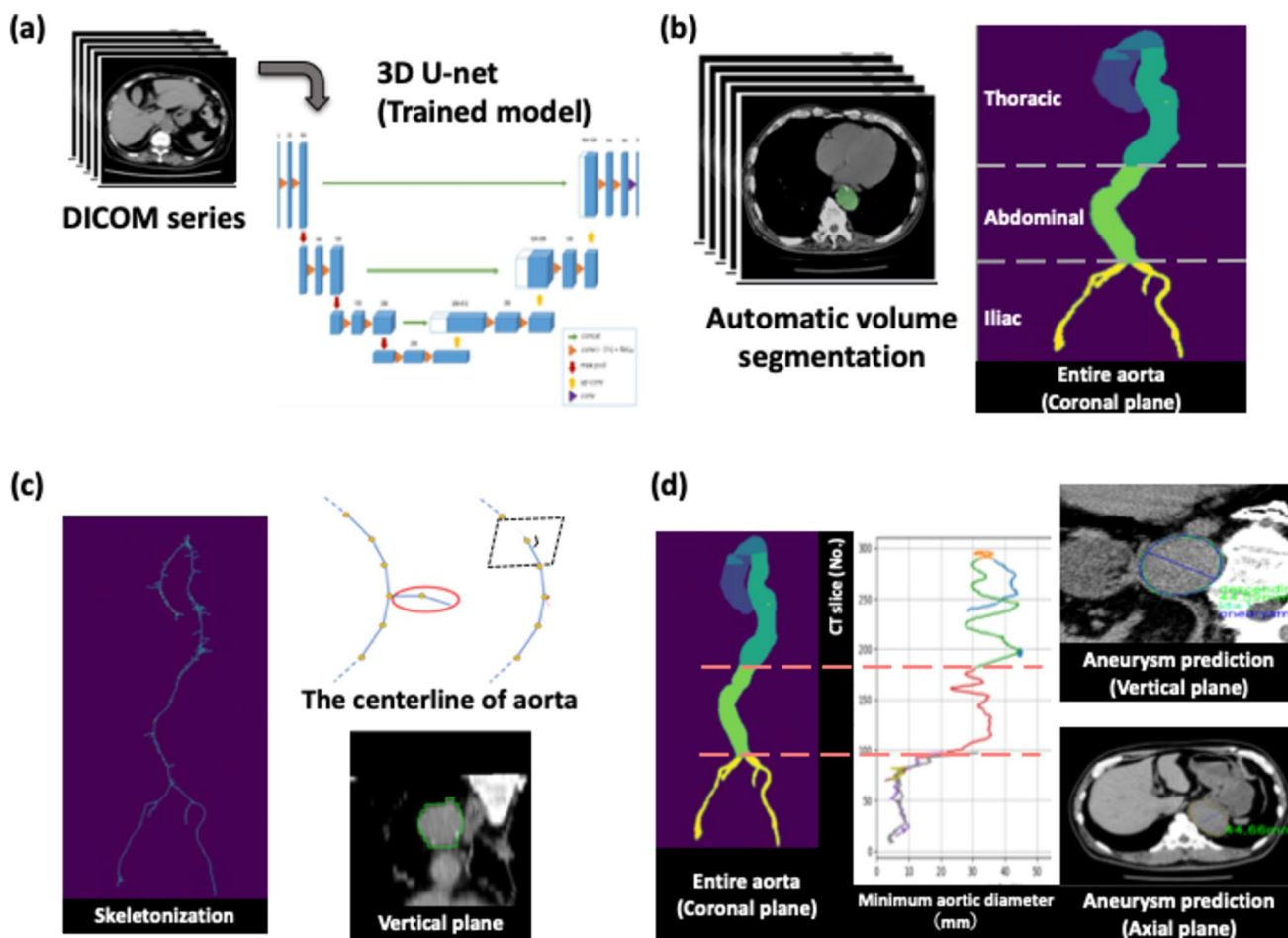


Fig. 1 Fully automatic workflow using the DLAI architecture. (a). The DICOM data with axial series were used as input to the segmentation model based on the 3D U-Net architecture. (b). Automatic volume segmentation of the entire aorta from the selected series and site segmentation. (c). The aortic centerline was determined using 3D skeletoniza-

tion and drawing the perpendicular view to the centerline. (d). The curve plotted with the minimum aortic diameter of the entire aorta and prediction of aortic aneurysms using peak detection algorithms. DLAI, deep learning-based artificial intelligence, DICOM, Digital Imaging and Communications in Medicine

was calculated using several points close to the centerline, and a section perpendicular to that line was created. This was defined as the perpendicular section of the aorta, and the same procedure was applied throughout the aorta. An ellipse was fitted around the perpendicular section at each coordinate point and adjusted to capture the minimum aortic diameter within the region, which was defined as the aortic minor diameter [17, 18]. The minimum aortic diameter was plotted every 2 mm along the long axis, and a curve representing the entire aortic diameter was generated. Using a peak detection algorithm and threshold diameters set for each aortic segment (≥ 45 mm for the thoracic, ≥ 30 mm for the abdominal, and ≥ 20 mm for the iliac regions), aneurysms were detected, and the maximum aortic diameter of the aneurysm was calculated [1, 2]. If multiple aneurysms were found in one series and they were ≤ 15 mm apart from each other on the horizontal slice, the largest one was kept. If they were > 15 mm apart, they were considered separate aneurysms. Finally, a visual interface of the segmented

aortic volume was generated using maximum intensity projection and stretched curved planar reformation. On the interface, the horizontal section closest to the center point of the vertical cross-section where the aneurysm was detected is displayed. The maximum aortic diameter of the aneurysm was also shown and automatically tagged (Fig. 2).

Validation of the DLAI system's accuracy

The diagnostic accuracy of the DLAI system for aortic aneurysms was evaluated using Dataset B. First, the accuracy of the DLAI results was assessed by referring to the radiology reports. A radiologist (Y.K) with 30 years of experience in cardiovascular imaging performed all measurements of the aortic aneurysms in the radiology reports. Based on the radiology reports, the maximum aortic diameter slice of the aortic aneurysm was determined (ground truth). If the aneurysm detected by the model was within 15 mm above or below the axial plane, it was defined as true positive (TP).

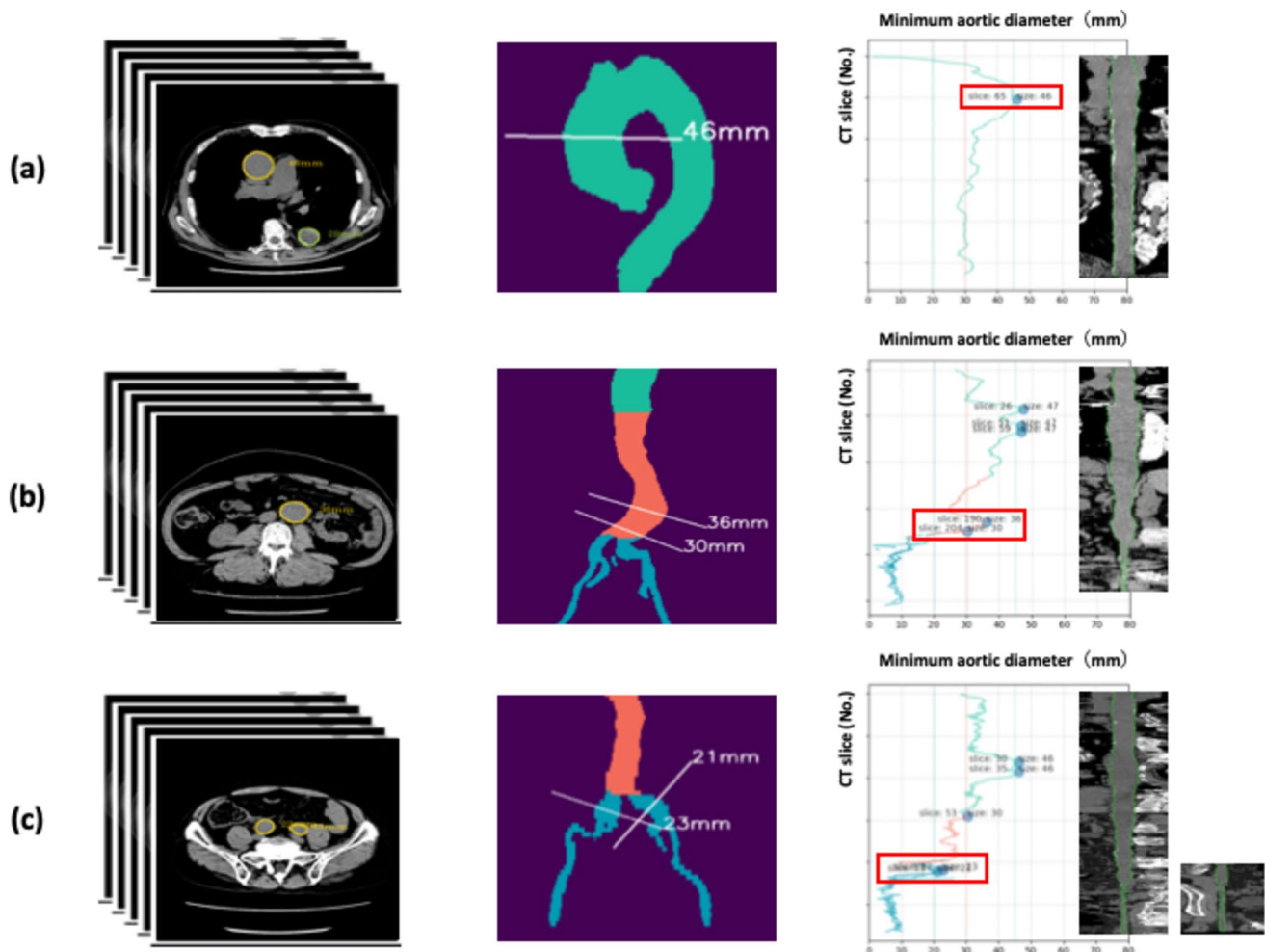


Fig. 2 Interface of detected aortic aneurysms using DLAI Axial (left) and coronal plane (middle). The curve plotted with the minimum aortic diameter and prediction of aortic aneurysms using peak detection

algorithms (right) in patients with thoracic (a), abdominal (b), and iliac (c) aortic aneurysms. DLAI, deep learning-based artificial intelligence

Using the AI results as a reference, the radiologist (Y.K.) then examined the CT images again and modified the ground truth data as necessary (modified ground truth; AI-assisted report). Finally, this was used to validate the final accuracy.

Statistical analysis

All statistical analyses were performed using SPSS® 28.0 (IBM Corporation, NY, USA). Continuous variables are presented as means $\pm$ standard deviations (SDs) and inter-quartile range as appropriate, and categorical variables are expressed as numbers and percentages. For Dataset A, the aortic segmentation quality was evaluated using the Dice similarity coefficient. The accuracy of aneurysm prediction was assessed using the classification metrics of sensitivity, precision, and F-measure (calculated as $(2 \times \text{sensitivity} \times \text{precision}) / (\text{sensitivity} + \text{precision})$). Inter-method and inter-reader agreements (Dataset B) were evaluated using the intraclass correlation coefficient (ICC) for each aortic segment. The correlation of the maximum diameter of the aneurysm between the DLAI system and AI-assisted reports was analyzed using Spearman's rank correlation coefficient.

Table 2. Comparison of DLAI model and radiology reports

	DLAI model vs. radiology reports	DLAI model vs. AI-assisted reports	Dif- fer- ence
Confusion matrix			
TP, n	198	209	11
FP, n	41	30	-11
FN, n	46	43	-3
Total			
Number of aneu- rysms, n	244	252	8
Sensitivity	0.81 (0.76–0.86)	0.83 (0.78–0.87)	0.02
Precision	0.83 (0.78–0.87)	0.87 (0.83–0.91)	0.04
F-measure	0.82	0.85	0.03
Thoracic			
Number of aneu- rysms, n	81	79	-2
Sensitivity	0.79 (0.69–0.87)	0.81 (0.71–0.89)	0.02
Precision	0.90 (0.81–0.96)	0.90 (0.81–0.96)	0
F-measure	0.84	0.85	0.01
Abdomen			
Number of aneu- rysms, n	127	132	5
Sensitivity	0.90 (0.83–0.94)	0.90 (0.84–0.95)	0.00
Precision	0.82 (0.75–0.88)	0.86 (0.79–0.91)	0.04
F-measure	0.86	0.88	0.02
Iliac			
Number of aneu- rysms, n	36	41	5
Sensitivity	0.56 (0.38–0.72)	0.63 (0.47–0.78)	0.07
Precision	0.69 (0.49–0.85)	0.90 (0.73–0.98)	0.21
F-measure	0.62	0.74	0.12

DLAI, deep learning-based artificial intelligence.

The Bland–Altman analysis method was performed to assess the interchangeability between the DLAI system and AI-assisted reports, bias, and 95% limits of agreement, which were calculated by multiplying the SD by ± 1.96 . The paired parametric Student's *t*-test was used for the significance analysis. *P*-values < 0.05 were used to denote statistical significance.

Results

The Dice scores between the predicted and manual reference segmentations using Dataset A were 0.90, 0.94, 0.93, and 0.84 for the entire aorta, thoracic aorta, abdominal aorta, and iliac artery, respectively. Figure 2 and Supplementary Movie S1 represent representative cases of aneurysms of the thoracic, abdominal, and iliac regions. Supplementary Movie S2 presents a comparison of the maximum aortic diameter of abdominal aortic aneurysms between the last and current non-contrast CT images using the DLAI system.

(Table 2). presents the performance of the DLAI system for the detection of fusiform aortic aneurysms. Based on the diameters from the radiology reports (ground truth), the diagnostic accuracy of the DLAI system for detecting fusiform aortic aneurysms was 0.81 for sensitivity, 0.83 for precision, and 0.82 for F-measure for the entire aorta. The accuracy was higher for the thoracic and abdominal aortae than for the iliac artery (Table 2). The reasons for detecting false cases using the DLAI system were overestimation or underestimation of the vessel diameter ($n=34$), detection of multiple peaks in a single aneurysm ($n=18$), failure to extract aortic region segments because of vessel tortuosity ($n=9$), failure to determine the position of the aorta ($n=7$), and inability to detect the peak due to the algorithm ($n=5$). For the group with a slice thickness of ≤ 3 mm, the accuracy of the DLAI system for the entire aorta was 0.81 for sensitivity, 0.91 for precision, and 0.85 for F-measure. For the group with a slice thickness of ≥ 4 mm, the accuracy of the DLAI system for the entire aorta was 0.84 for sensitivity, 0.86 for precision, and 0.85 for F-measure.

The radiologist (Y.K.) re-read the CT images using the DLAI results as a reference, and 11 (4.4%) additional aneurysms were identified, with five in the abdominal region and six in the iliac region, and three did not meet the criteria for aortic aneurysms. The accuracy compared with AI-assisted reports (modified ground truth) increased to 0.83 for sensitivity, 0.87 for precision, and 0.85 for F-measure. The improvement in accuracy for IAA imaging was particularly significant (Table 2).

For the 209 TP cases, the difference in the size of the fusiform aortic aneurysms between the DLAI system and AI-assisted reports was 0.86 ± 2.72 mm, with a strong

Table 3. Comparison of the DLAI system and radiology reports

	All	Thoracic	Abdomen	Iliac
Number of aneurysms, n	209	64	119	26
Average of AI-assisted reports, mm	43.4±11.0	51.8±6.5	42.8±8.8	25.5±3.9
Average of DLAI, mm	42.6±10.6	50.4±5.1	42.3±8.6	24.4±4.9
Mean difference, mm	0.86±2.72	1.33±2.65	0.55±2.64	1.13±3.00
Spearman's coefficient	0.969	0.923	0.954	0.781
P-value	<0.01	<0.01	<0.01	<0.01

DLAI, deep learning-based artificial intelligence.

correlation (ICC=0.97) (Table 3 and Fig. 3). However, the average of the AI-assisted reports (43.4±11.0 mm) was significantly higher than that of the DLAI system (42.6±10.6 mm, $p<0.01$) (Table 3).

Discussion

To the best of our knowledge, this study is the first to introduce a DLAI system for detecting fusiform aortic aneurysms and measuring the diameter of the entire aorta on non-contrast CT images. The cases in both the training and validation datasets showed higher median Agatston scores in the thoracic, abdominal, and iliac regions compared with prior studies, indicating more severe aortic calcification [19, 20]. Despite such intricate anatomical variations, the segmentation performance for the entire aortic volume remained consistent, with a median Dice coefficient of 0.90. The mean absolute error between the automatically measured and reader-measured diameters was ≤ 1.5 mm for the thoracic, abdominal, and iliac arteries. The improvement in the detection accuracy for IAAs was particularly significant after rereading the CT images using the DLAI results as a reference. Thus, a more accurate DLAI approach aims not only to detect and measure fusiform aneurysms on non-contrast CT images but also to improve diagnostic accuracy.

Because aortic aneurysm screening is typically not routinely performed and considering the asymptomatic nature and high mortality rate associated with aortic rupture, all

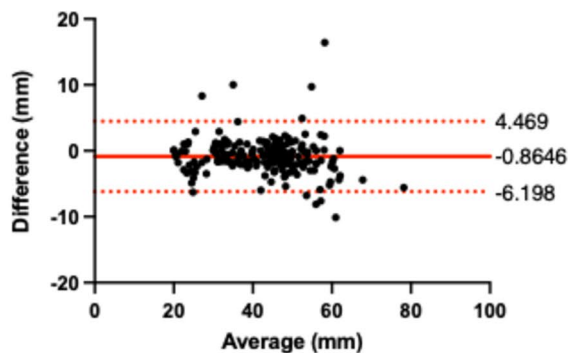
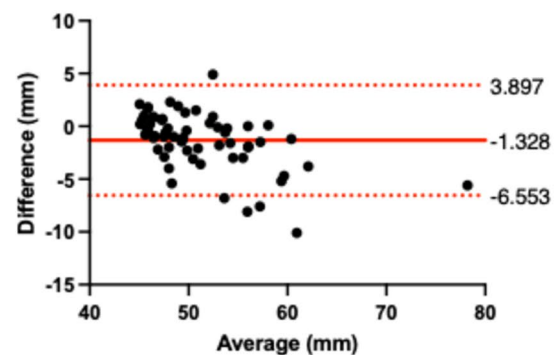
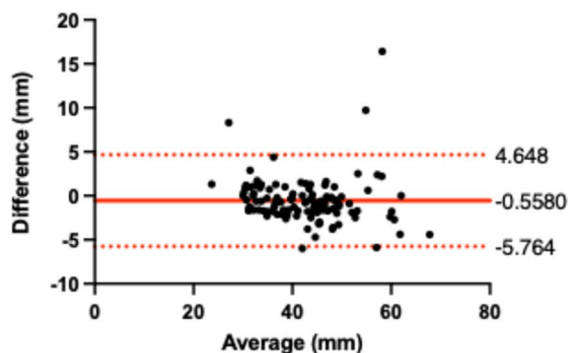
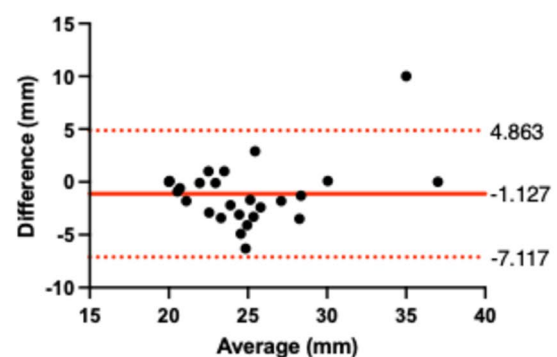
DLAI system vs. AI-assisted reports for entire aorta**DLAI system vs. AI-assisted reports for thoracic aorta****DLAI system vs. AI-assisted reports for abdominal aorta****DLAI system vs. AI-assisted reports for iliac artery**

Fig. 3 Bland–Altman plots of the difference in diameters between the DLAI system and AI-assisted reports for the entire aorta, thoracic aorta, abdominal aorta, and iliac artery. The graphs show the relation-

ships between the means and differences for each functional parameter. Solid line, bias; dotted lines, upper and lower 95% limits of agreement. DLAI, deep learning-based artificial intelligence

available studies, including non-contrast CT imaging, should be used for their diagnosis [1–3, 21]. In contrast, assessment of the aortic diameter becomes more challenging in nondedicated examinations. Phenomena, such as increased workload in radiology, can contribute to missed diagnosis of aortic aneurysms [22, 23]. However, it remains unclear whether radiologists evaluate and report cardiovascular abnormalities from nonvascular chest CT images, with disclosed rates being as high as 63% for unreported cardiovascular findings from chest CT performed for pulmonary evaluation, remains unclear [24]. Moreover, the shortcomings of traditional CT-based manual measurement include difficulties in accurately detecting the maximum diameter across multiple slices and three views, inadequate reproducibility, prolonged measurement time, and subjective determination of individuals [4, 5].

DL offers a pathway for scalability and is primed to refine and elevate performance metrics because its widespread use provides algorithms with increased and diversified data inputs [6–9]. DL is an advanced AI technique that has been applied to measuring aortic diameters according to current guidelines and has shown good results in cases without aneurysms [25, 26]. However, the performance of AI applications should be compared to the current clinical practice in a routine clinical setting to determine whether DLAI systems for aortic evaluation can entirely replace human reading. This would indicate sufficient reliability for clinical implementation if DLAI system performance is similar to that of manual aortic measurements and if there is variability between human readers. Several DLAI tools have been explored for measuring aortic diameter using CTA [10, 18, 27]. A recent study reported the reliability of a fully automated AI-driven solution capable of quick aortic segmentation and analysis of the diameter and volume of each segment using CTA images [10]. However, reports of DLAI systems for detecting fusiform aortic aneurysms are limited, and guideline-compliant measurements of the diameter of the entire aorta in non-contrast CT images have been rarely performed [11–13]. Moreover, no studies have validated its diagnostic accuracy in clinical settings.

The capacity of our DLAI system to measure the size of fusiform aortic aneurysms remained within the clinically accepted error range. Our comprehensive analysis of the entire aorta achieved a median Dice coefficient of 0.90, matching previously published data [10, 27]. Bland–Altman analysis further validated these findings, displaying nearly no bias. Although the systematic diameter tended to be underestimated by the DLAI system, the discrepancy between conventional and DLAI-assisted readings did not exceed 1.5 mm. The development of advanced algorithms is necessary to improve accuracy; however, we demonstrated that DLAI is a promising tool for improving the

time efficiency of radiological reporting and therefore may support the application of guideline-compliant aorta assessments in clinical settings. In this study, the accuracy of DLAI in identifying and measuring aortic aneurysms was evaluated using non-contrast CT images. However, CTA remains the gold standard for aortic aneurysm measurement. Future studies directly comparing DLAI measurements with CTA-based measurements are necessary to further validate the clinical utility of the DLAI system.

In this study, the diagnostic accuracy of the proposed DLAI system was lower for the iliac artery than for the thoracic and abdominal aortae. In contrast, the improvement in diagnostic accuracy for IAAs was particularly significant after rereading of the CT images using the DLAI results as a reference by the expert radiologist. IAAs are rare conditions, with a prevalence of 0.03% in the general population, comprising 2%–7% of all abdominal aneurysms [21, 28, 29]. Moreover, IAAs typically remain asymptomatic and are difficult to discover because of their deep location within the pelvic cavity [21]. Rupture is a common initial manifestation of IAA and is associated with significant mortality rates [21, 30]. Although CTA has higher diagnostic accuracy and provides detailed anatomical images of the affected arteries, IAAs are often incidentally discovered because of the increased use of non-contrast CT scans [21]. Moreover, radiologists do not prioritize diagnosing subtle cardiovascular abnormalities, including IAAs, in non-contrast CT scans [23]. The results of our study also revealed that DLAI assistance can significantly influence a radiologist's decision-making; however, radiologists can actively reject or correct proposed DLAI measurements, which should be critically discussed with a focus on the possible risks of AI applications. Therefore, our DLAI system plays a crucial role in detecting and assessing aortic aneurysms, particularly for IAAs.

Our DLAI system was designed to run silently in the background of PACS. Furthermore, it allows the segmentation of the aorta in the selected series and the measurement of the maximum diameters and provides visual outputs of the segmented aortic lesions (Fig. 2). We developed an adjunctive DICOM axial series comprising an aortic overlay (Supplementary Movies S1 and S2). Thus, our DLAI system was developed to help radiologists, not to replace them. These supplementary series enable radiologists to assess segmentation quality quickly and accurately. We also presented the potential benefits of follow-up scans (Supplementary Movie S2). The evaluation of such examinations requires consistent measurement of the aortic diameter along the aorta, particularly focusing on the positioning and angulation of the diameter measurements. The detection model developed in this study and the dataset used have the potential to be made publicly available in the future.

Limitations

This study has several limitations that should be considered. First, the dataset was obtained from a single institution only; therefore, validation in multiple institutions is required in future studies. Furthermore, external validation in this study was conducted using data from a same institution and therefore limit the model's generalizability. Future studies utilizing publicly available datasets are needed to improve the model's generalizability. Second, manual segmentation was used during the model development stage in this study. Recent foundation models such as MedSAM and VesselFM utilize large-scale and diverse training datasets to enable semi-automatic segmentation and demonstrate superior generalization performance [31, 32]. These models have achieved high efficiency and reproducibility results. Although no foundation model-based approach for medical image segmentation existed when we developed this model, integration of such models could further enhance our system's performance more efficiently, particularly in the context of external validation. Third, the study was retrospective in nature and did not evaluate the negative predictive value for cases without aortic aneurysms, thereby making the validation of system accuracy insufficient. Furthermore, the numbers of cases of thoracic, abdominal, and iliac aneurysms used for model training were not balanced. In addition, the proportion of male patients was higher in both the model development dataset and the validation dataset compared to the 44% to 66% reported in previous studies [17, 18]. Fourth, readings from only one expert were used as a reference for all measurement comparisons. High variability can also exist between experts. Finally, further prospective studies are necessary to determine whether our DLAI system can prevent radiologists from overlooking aneurysms on CT performed for nonvascular investigations, particularly subtle ones.

Conclusions

Our DLAI system was highly accurate in detecting fusiform aneurysms of the entire aorta and measuring the maximum aortic diameter on non-contrast CT images. The study results suggest that a more accurate DLAI approach aims not only to reduce labor-intensive, time-consuming, and repetitive manual tasks but also to improve diagnostic accuracy.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10554-026-03630-y>.

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Author contributions Jumpei Fujiwara: Writing – original draft, Investigation, Data curation, Conceptualization. Makoto Orii: Writing – review & editing, Investigation, Data curation. Kohei Oyamada, Kenta Araki, Masahiko Ogura, Tomoyasu Ito, Daria Bunu, Ryoto Ando, and Takashi Okada: Writing – review & editing, Software, Methodology, Investigation. Kunihiro Yoshioka: Writing – review & editing, Supervision, Investigation.

No datasets were generated or analysed during the current study.

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Declarations

Competing interests The authors declare no competing interests.

Ethical The study was approved by the institution's human research committee (Iwate Medical University).

References

- Isselbacher EM, Preventza O et al (2022) Peer review committee members 2022 ACC/AHA guideline for the diagnosis and management of aortic disease: a report of the american heart association/american college of cardiology joint committee on clinical practice guidelines. *Circulation* 146:e334–e482
- Erbel R, Aboyans V et al (2014) ESC Committee for Practice Guidelines. 2014 ESC Guidelines on the diagnosis and treatment of aortic diseases: Document covering acute and chronic aortic diseases of the thoracic and abdominal aorta of the adult. The Task Force for the Diagnosis and Treatment of Aortic Diseases of the European Society of Cardiology (ESC). *Eur Heart J* 35:2873–2926
- Isselbacher EM (2005) Thoracic and abdominal aortic aneurysms. *Circulation* 111:816–828
- Mora CE, Marcus CD, Barbe CM, Ecarnot FB, Long AL (2014) Measurement of maximum diameter of native abdominal aortic aneurysm by angio-CT: reproducibility is better with the semi-automated method. *Eur J Vasc Endovasc Surg* 47:139e150
- Mora CE, Marcus CD, Barbe CM, Ecarnot FB, Long AL (2015) Maximum diameter of native abdominal aortic aneurysm measured by angio-computed tomography: reproducibility and lack of consensus impacts on clinical decisions. *Aorta Stamford* 3:47–55
- Jiang Y, Yang M, Wang S, Li X, Sun Y (2020) Emerging role of deep learning-based artificial intelligence in tumor pathology. *Cancer Commun* 40:154–166
- Chi W, Ma L, Wu J, Chen M, Lu W, Gu X (2020) Deep learning-based medical image segmentation with limited labels. *Phys Med Biol*. <https://doi.org/10.1088/1361-6560/abc363>
- Wang G, Li W, Zuluaga MA, Pratt R, Patel PA, Aertsen M, Doel T, David AL, Deprest J, Ourselin S, Vercauteren T (2018) Interactive medical image segmentation using deep learning with image-specific fine tuning. *IEEE Trans Med Imaging* 37:1562–1573
- Cai L, Gao J, Zhao D (2020) A review of the application of deep learning in medical image classification and segmentation. *Ann Transl Med* 8:713
- Postiglione TJ, Guillo E, Heraud A, Rossillon A, Bartoli M, Herpe G, Adam C, Fabre D, Ardon R, Azarine A, Haulon S (2024) Multicentric clinical evaluation of a computed tomography-based fully automated deep neural network for aortic maximum diameter and volumetric measurements. *J Vasc Surg* 79:1390–1400
- Hamelink II, de Hei EEJ, Pelgrim GJGJ, Kwee TCT, van Ooijen PMA, de Bock GHT, Vliegenthart RR (2023) Validation of an

- AI-based algorithm for measurement of the thoracic aortic diameter in low-dose chest CT. *Eur J Radiol* 167:111067
12. Lo Piccolo F, Hinck D, Segeroth M, Sperl J, Cyriac J, Yang S, Rapaka S, Bremerich J, Sauter AW, Pradella M (2023) Impact of retraining a deep learning algorithm for improving guideline-compliant aortic diameter measurements on non-gated chest CT. *Eur J Radiol* 168:111093
 13. Rueckel J, Reidler P, Fink N, Sperl J, Geyer T, Fabritius MP, Ricke J, Ingrisich M, Sabel BO (2021) Artificial intelligence assistance improves reporting efficiency of thoracic aortic aneurysm CT follow-up. *Eur J Radiol* 134:109424
 14. Agatston AS, Janowitz WR, Hildner FJ, Zusmer NR, Viamonte M, Detrano R (1990) Quantification of coronary artery calcium using ultrafast computed tomography. *J Am Coll Cardiol* 15:827–832
 15. Cardoso MJ, Li W, Brown R, Ma N, Kerfoot E, Wang Y, Murrey B, Myronenko A, Zhao C, Yang D, Nath V, He Y, Xu Z, Hatamizadeh A, Myronenko A, Zhu W, Liu Y, Zheng M, Tang Y, Yang I, Zephyr M, Hashemian B, Alle S, Darestani MZ, Budd C, Modat M, Vercauteren T, Wang G, Li Y, Hu Y, Fu Y, Gorman B, Johnson H, Genereaux B, Erdal BS, Gupta V, Diaz-Pinto A, Dourson A, Maier-Hein L, Jaeger PF, Baumgartner M, Kalpathy-Cramer J, Flores M, Kirby J, Cooper L, A.D., Roth HR, Xu D, Bericat D, Floca R, Zhou SK, Shuaib H, Farahani K, Maier-Hein KH, Aylward S, Dogra P, Ourselin S, Feng A. 2022 Monai: An open-source framework for deep learning in healthcare. *arXiv*. 2211.02701.
 16. Halkoaho J, Niiranen O, Salli E, Kaseva T, Savolainen S, Kangasniemi M, Hakovirta H (2024) Quantifying the calcification of abdominal aorta and major side branches with deep learning. *Clin Radiol* 79:e665–e674
 17. Lu J-T, Brooks R, Hahn S, Chen J, Buch V, Kotecha G, Andriole KP, Ghoshhajra B, Pinto J, Vozila P, Michalski M, Tenenholtz NA (2019) DeepAAA: clinically applicable and generalizable detection of abdominal aortic aneurysm using deep learning. *MICCAI; LNCS*. 11765:723–731
 18. Macruz FBC, Lu C, Strout J, Takigami A, Brooks R, Doyle S, Yun M, Buch V, Hedgire S, Ghoshhajra B (2022) Quantification of the thoracic aorta and detection of aneurysm at CT: development and validation of a fully automatic methodology. *Radiol Artif Intell* 4:e210076
 19. Razavi AC, Dzaye O, Cainzos-Achirica M, Dardari Z, Van Assen M, Quyyumi AA, Nasir K, Carr JJ, Budoff MJ, Blumenthal RS, Raggi P, De Cecco CN, Sperling LS, Blaha MJ, Whelton SP (2025) Association of thoracic aortic calcium with incident cardiovascular disease and all-cause mortality across the spectrum of coronary artery calcium burden. *Am J Prev Cardiol* 21:100916
 20. Jurgens PT, Carr JJ, Terry JG, Rana JS, Jacobs DR Jr, Duprez DA (2021) Association of abdominal aorta calcium and coronary artery calcium with incident cardiovascular and coronary heart disease events in Black and White middle-aged people: the Coronary Artery Risk Development in Young Adults Study. *J Am Heart Assoc* 10:e023037
 21. Kotsis T, Louizos LA, Pappas E, Theodoraki K (2014) Complex common and internal iliac or aortoiliac aneurysms and current approach: individualised open-endovascular or combined procedures. *Int J Vasc Med* 2014:178610
 22. Alexander R, Waite S, Bruno MA, Krupinski EA, Berlin L, Macknik S, Martinez-Conde S (2022) Mandating limits on workload, duty, and speed in radiology. *Radiology* 304:274–282
 23. Krueger M, Cronin P, Sayyoush M, Kelly AM (2019) Significant incidental cardiac disease on thoracic CT: what the general radiologist needs to know. *Insights Imaging* 10:10
 24. Sverzellati N, Arcadi T et al (2016) Under-reporting of cardiovascular findings on chest CT. *Radiol Med* 121:190–199
 25. Monti CB, van Assen M, Stillman AE, Lee SJ, Hoelzer P, Fung GSK, Secchi F, Sardanelli F, De Cecco CN (2022) Evaluating the performance of a convolutional neural network algorithm for measuring thoracic aortic diameters in a heterogeneous population. *Radiol Artif Intell* 4:e210196
 26. Pradella M, Achermann R, Sperl JI, Kärgel R, Rapaka S, Cyriac J, Yang S, Sommer G, Stieltjes B, Bremerich J, Brantner P, Sauter AW (2022) Performance of a deep learning tool to detect missed aortic dilatation in a large chest CT cohort. *Front Cardiovasc Med* 9:972512
 27. Kim T, On S, Gwon JG, Kim N (2024) Computed tomography-based automated measurement of abdominal aortic aneurysm using semantic segmentation with active learning. *Sci Rep* 14:8924
 28. Johnston KW, Rutherford RB, Tilson MD, Shah DM, Hollier L, Stanley JC (1991) Suggested standards for reporting on arterial aneurysms. Subcommittee on Reporting Standards for Arterial Aneurysms, Ad Hoc Committee on Reporting Standards, Society for Vascular Surgery and North American Chapter, International Society for Cardiovascular Surgery. *J Vasc Surg* 13:452–458
 29. Krupski WC, Selzman CH, Florida R, Strecker PK, Nehler MR, Whitehill TA (1998) Contemporary management of isolated iliac aneurysms. *J Vasc* 28:1–13
 30. Melas N, Saratzis A, Dixon H, Saratzis N, Lazaridis J, Perdikides T, Kiskinis D (2011) Isolated common iliac artery aneurysms: a revised classification to assist endovascular repair. *J Endovasc Ther Off J Int Soc Endovasc Spec* 18:697–715
 31. Ma J, He Y, Li F, Han L, You C, Wang B (2024) Segment anything in medical images. *Nat Commun* 15:654
 32. Wittmann B, Wattenberg Y, Amiranashvili T, Shit S, Menze B, vesselFM A. 2025. Foundation Model for Universal 3D Blood Vessel Segmentation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 20874–20884.

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